

Project Interim Report
On
LibraryManagement System
(Using Core Java & OOP Concepts)
RU-100-01-00012
Object Oriented Programming With Java
SESSION: 2025-26



Submitted By
Mohammad
Ayan
RU-25-10828

Bachelor of Technology in Computer Science &
Engineering in association with Google

Under the Guidance of

Mr. Soumik Karmakar

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
RUNGTA INTERNATIONAL SKILLS UNIVERSITY

BHILAI , CG

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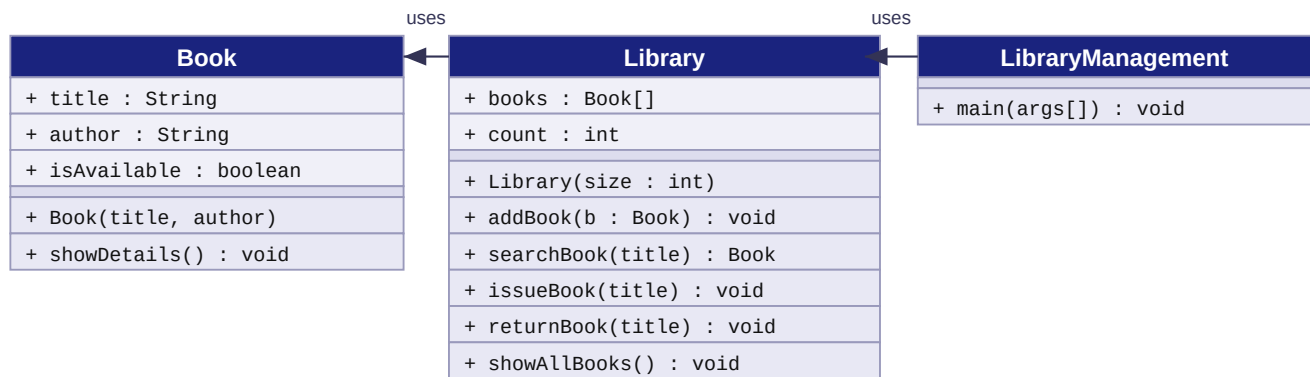
1. About the Project

A menu-driven Library Management System built in Java using core Object-Oriented Programming concepts. The program runs in a loop and lets the user add books, search them, issue and return them — just like a real library counter. All classes are written inside a single **LibraryManagement.java** file for simplicity.

2. Program Structure

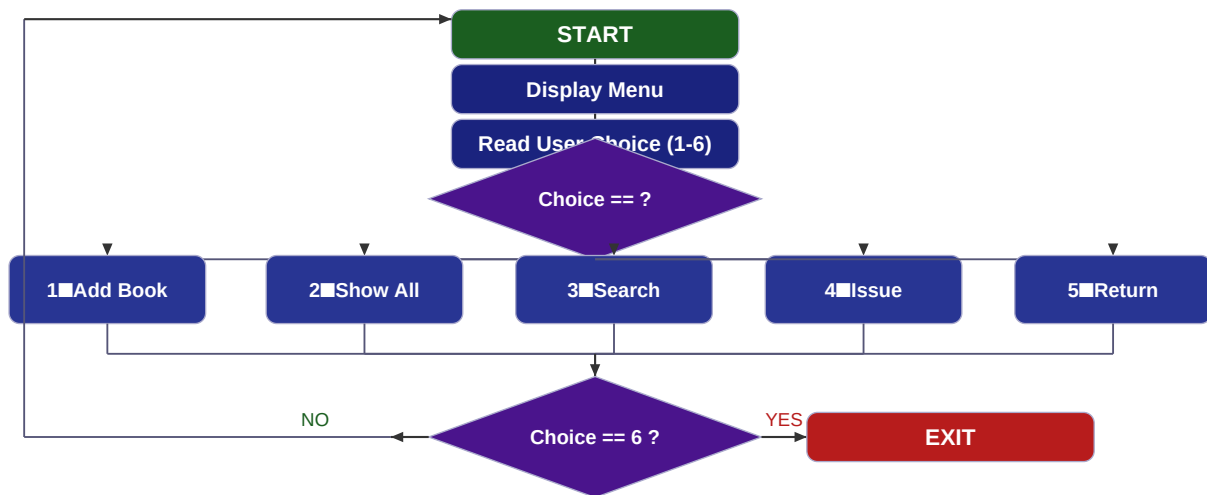
Class	Field / Method	Type	Description
Book	title	String	Name of the book
	author	String	Author's name
	isAvailable	boolean	true=Available, false=Issued
	Book(title, author)	Constructor	Initializes fields, sets availability=true
	showDetails()	void	Prints title, author, current status
	books[]	Book[]	Array storing all Book objects
Library	count	int	Tracks number of books added
	Library(size)	Constructor	Creates array with given capacity
	addBook(b)	void	Adds a Book to the array
	searchBook(title)	Book	Case-insensitive search, returns Book or null
	issueBook(title)	void	Issues book if available, else shows message
	returnBook(title)	void	Returns book, updates availability
LibraryManagement	showAllBooks()	void	Displays all books with status
	main(args[])	void	Scanner input + do-while menu loop

3. Class Diagram



The diagram above shows three classes. **LibraryManagement** uses **Library**, and **Library** uses **Book** through a `Book[]` array. Each class has its own fields and methods following Encapsulation.

4. Flowchart — Program Logic



The flowchart shows the do-while menu loop. The program displays the menu, reads the user's choice, executes the selected operation, and loops back until the user enters 6 (Exit).

5. OOP Pillars Used

Pillar	Used?	Where in Code
Encapsulation	Yes	Data and methods are bundled inside Book and Library classes
Abstraction	Yes	User calls issueBook() — internal search/update logic is hidden
Inheritance	No	Not required for this scope (would apply if EBook extended Book)
Polymorphism	Partial	showDetails() called on different Book objects — each prints its own data

6. Features

- Empty library at start — no hardcoded books
- User adds books manually at runtime via Scanner
- Search book by title — case-insensitive matching
- Issue a book — marks status as Issued
- Return a book — marks status back to Available
- View all books with their current availability status
- Handles all edge cases: book not found, already issued, not issued
- Menu runs in a do-while loop until user selects Exit (6)

7. How to Run

Step 1 — Compile:

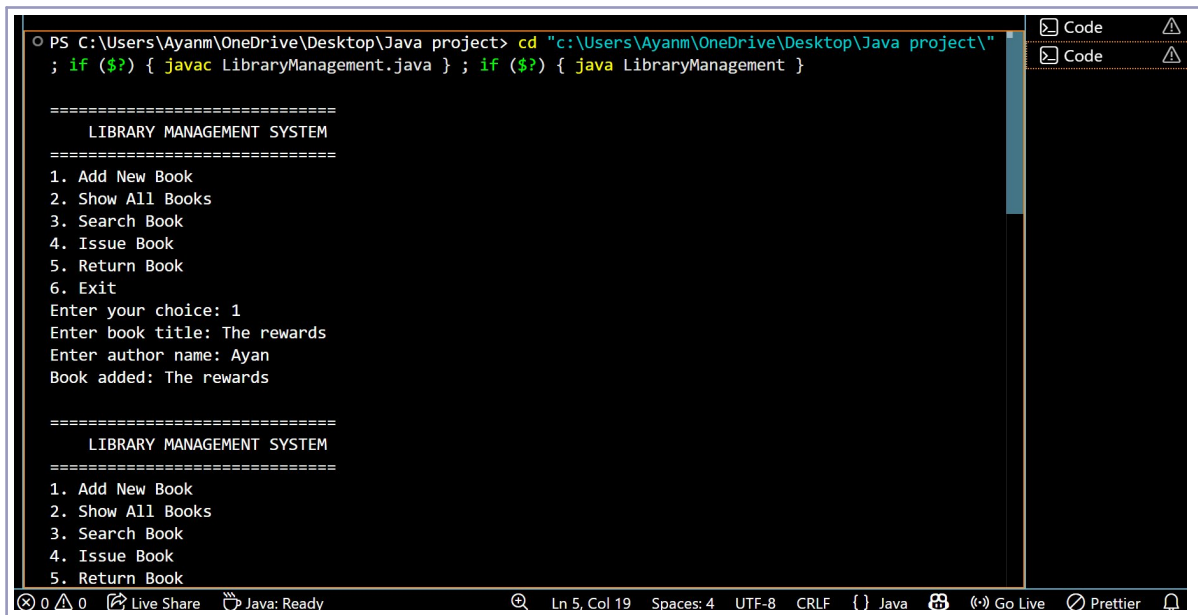
```
javac LibraryManagement.java
```

Step 2 — Run:

```
java LibraryManagement
```

Requires Java 8 or above. No external libraries needed.

8. Program Output — Screenshots

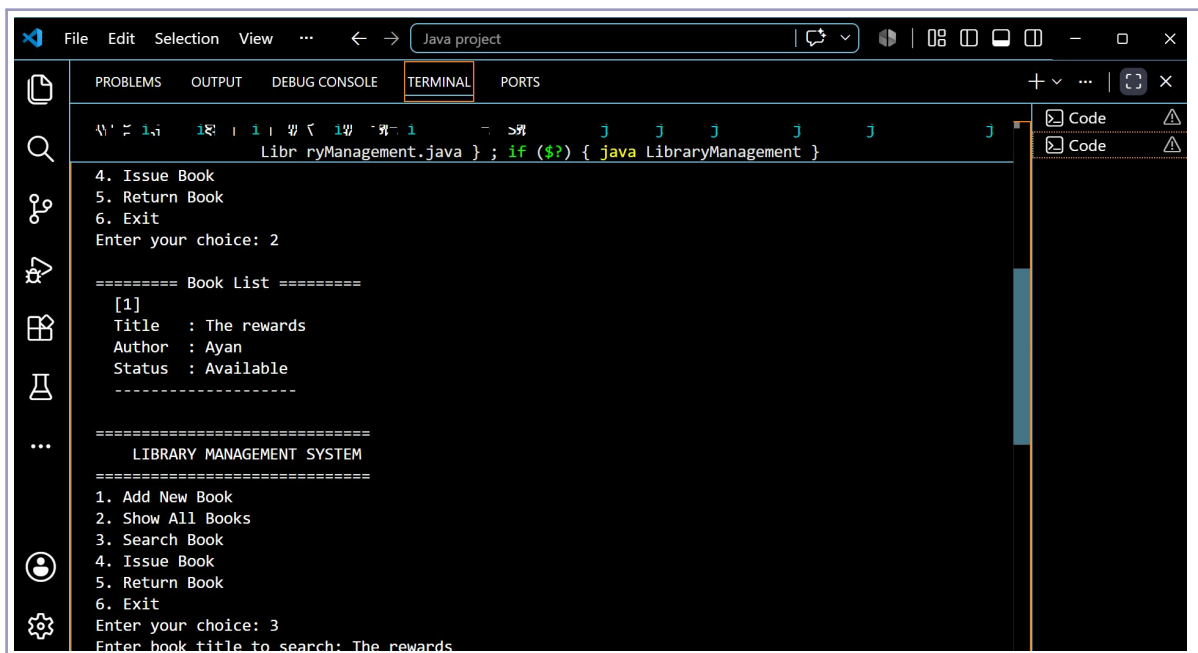


```
PS C:\Users\Ayanm\OneDrive\Desktop\Java project> cd "c:\Users\Ayanm\OneDrive\Desktop\Java project\"
; if ($?) { javac LibraryManagement.java } ; if ($?) { java LibraryManagement }

=====
LIBRARY MANAGEMENT SYSTEM
=====
1. Add New Book
2. Show All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter your choice: 1
Enter book title: The rewards
Enter author name: Ayan
Book added: The rewards

=====
LIBRARY MANAGEMENT SYSTEM
=====
1. Add New Book
2. Show All Books
3. Search Book
4. Issue Book
5. Return Book
```

Screenshot 1: Program starts, user adds book 'The rewards' by Ayan using Option 1



```
4. Issue Book
5. Return Book
6. Exit
Enter your choice: 2

===== Book List =====
[1]
Title : The rewards
Author : Ayan
Status : Available
-----

=====
LIBRARY MANAGEMENT SYSTEM
=====
1. Add New Book
2. Show All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter your choice: 3
Enter book title to search: The rewards
```

Screenshot 2: Option 2 — Show All Books, then Option 3 — Search Book

```
Libr ryManagement.java } ; if ($?) { java LibraryManagement }

6. Exit
Enter your choice: 3
Enter book title to search: The rewards

Book Found:
Title : The rewards
Author : Ayan
Status : Available
-----

=====
LIBRARY MANAGEMENT SYSTEM
=====
1. Add New Book
2. Show All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter your choice: 4
Enter book title to issue: The rewards
```

Screenshot 3: Search result shown, then Option 4 — Issue Book executed